

PROJECT SUMMARY

1) Project goal

Build an **AI-driven browser-based web platform MVP**:

- User registration and login
- Business registration
- Business verification
- User ratings and reviews
- AI-supported business search
- Admin/verifier dashboard

2) MVP tech stack

Core stack

- **Frontend + full-stack app:** Next.js (App Router) + React + TypeScript
- **Backend logic:** Next.js Route Handlers / Server Actions
- **Database:** PostgreSQL
- **ORM:** Prisma
- **Authentication:** Auth.js
- **Validation:** Zod
- **Forms:** React Hook Form
- **UI styling:** Tailwind CSS
- **Search:** PostgreSQL full-text search + pgvector
- **File storage:** S3-compatible object storage
- **Testing:** Playwright
- **Version control / CI:** GitHub + GitHub Actions
- **Deployment:** Vercel, Railway, Render, or Docker on a VPS

Why this stack

This stack is the best fit because:

- **Next.js App Router** supports layouts, routing, server/client components, and backend features in the same application.
- **Prisma** gives type-safe access to PostgreSQL and supports migrations cleanly.
- **Auth.js** is designed for web authentication across modern JavaScript frameworks, including Next.js.
- **pgvector** supports vector similarity search directly inside Postgres, including exact and approximate nearest-neighbor search.

Frontend/Web App

- Next.js
- TypeScript
- Tailwind CSS
- React Hook Form
- Zod

Backend

- Next.js Route Handlers / Server Actions
- Prisma ORM
- Auth.js

Database

- PostgreSQL
- pgvector

Storage

- S3-compatible object storage for documents and images

AI/Search

- embedding generation service
- PostgreSQL full-text search
- pgvector similarity search
- simple ranking layer

Testing/DevOps

- Playwright
- Docker
- GitHub Actions
- Vercel/Railway/Render

3) High-level system architecture

Architecture style

Use a **modular monolith** for the MVP.

- one main application
- one main database
- one file storage service
- optional external AI embedding provider
- admin and public features inside the same platform

4) Core modules for the MVP

Public module

- Homepage
- Search page
- Business listing page
- Business detail page

Authentication module

- Register
- Login
- Logout
- Password reset
- Email verification
- Session/role enforcement

Business module

- Register a business
- Edit business profile
- Upload business documents
- Track verification status

Verification module

- Admin queue
- Document review
- Approve/reject verification
- Add reason/comments
- Audit trail

Ratings and reviews module

- Create rating
- Create review
- Edit own review
- Admin moderation

AI-supported search module

- Keyword search
- Semantic search
- Suggested results
- Filtering and ranking

5) Database entities

Core entities

1. **User**
2. **Role**
3. **UserRole**
4. **Business**
5. **BusinessCategory**
6. **BusinessDocument**
7. **VerificationRequest**
8. **VerificationDecision**
9. **Rating**
10. **Review**
11. **ReviewReport**
12. **SearchEmbedding**
13. **AuditLog**
14. **Location**

